SyriaTel Churn Analysis

Ngundo Muithya - Senior Data Scientist

Project Overview

SyriaTel is interested in predicting customer churn

Create a predictive model that can reliably predict customer churn

Used customer data to build two models and selected the best

Give business recommendations to reduce churn

Business Understanding

Tasks

- Create a predictive model that can be used to predict customer churn
- Give recommendations on how to reduce churn

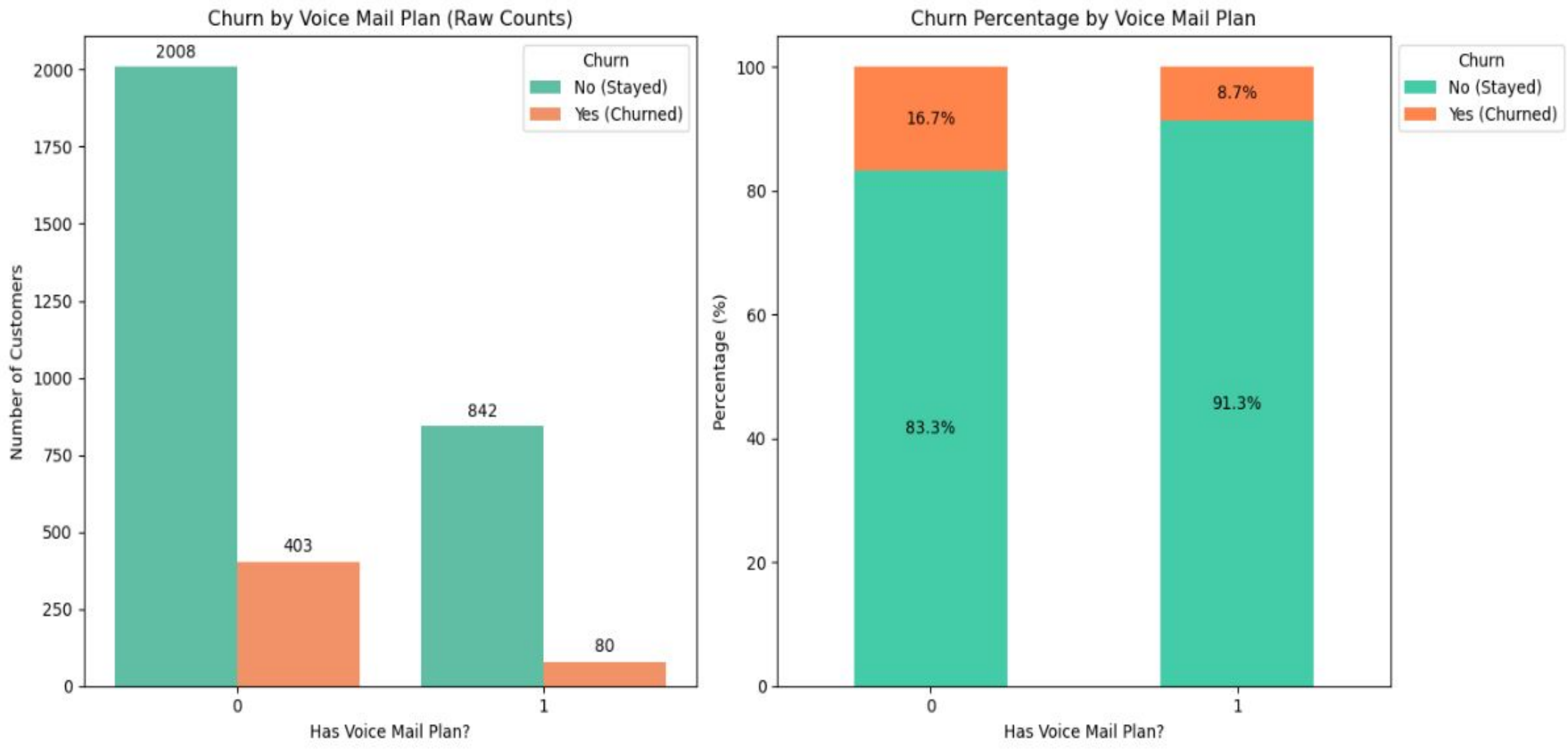
Deliverables

- Predictive model
- Recommendations to reduce churn

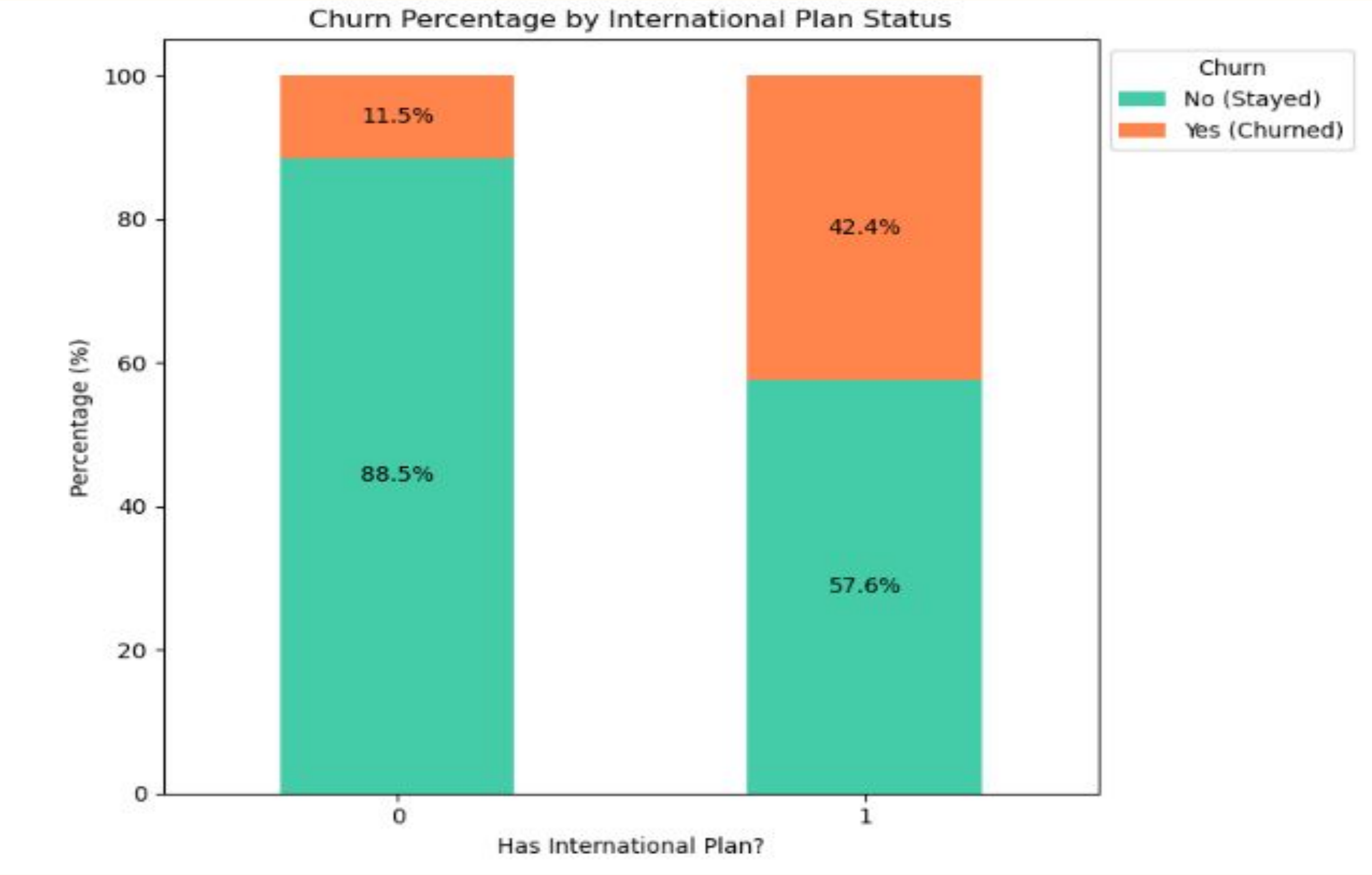
Data Understanding

- Used data on customer churn provided by SyriaTel
- Had 3333 entries
- Relevant metrics such as customer service calls and voicemail plan were used to predict churning

Churn distribution by voicemail plan



Churn distribution by international plan



Model Building

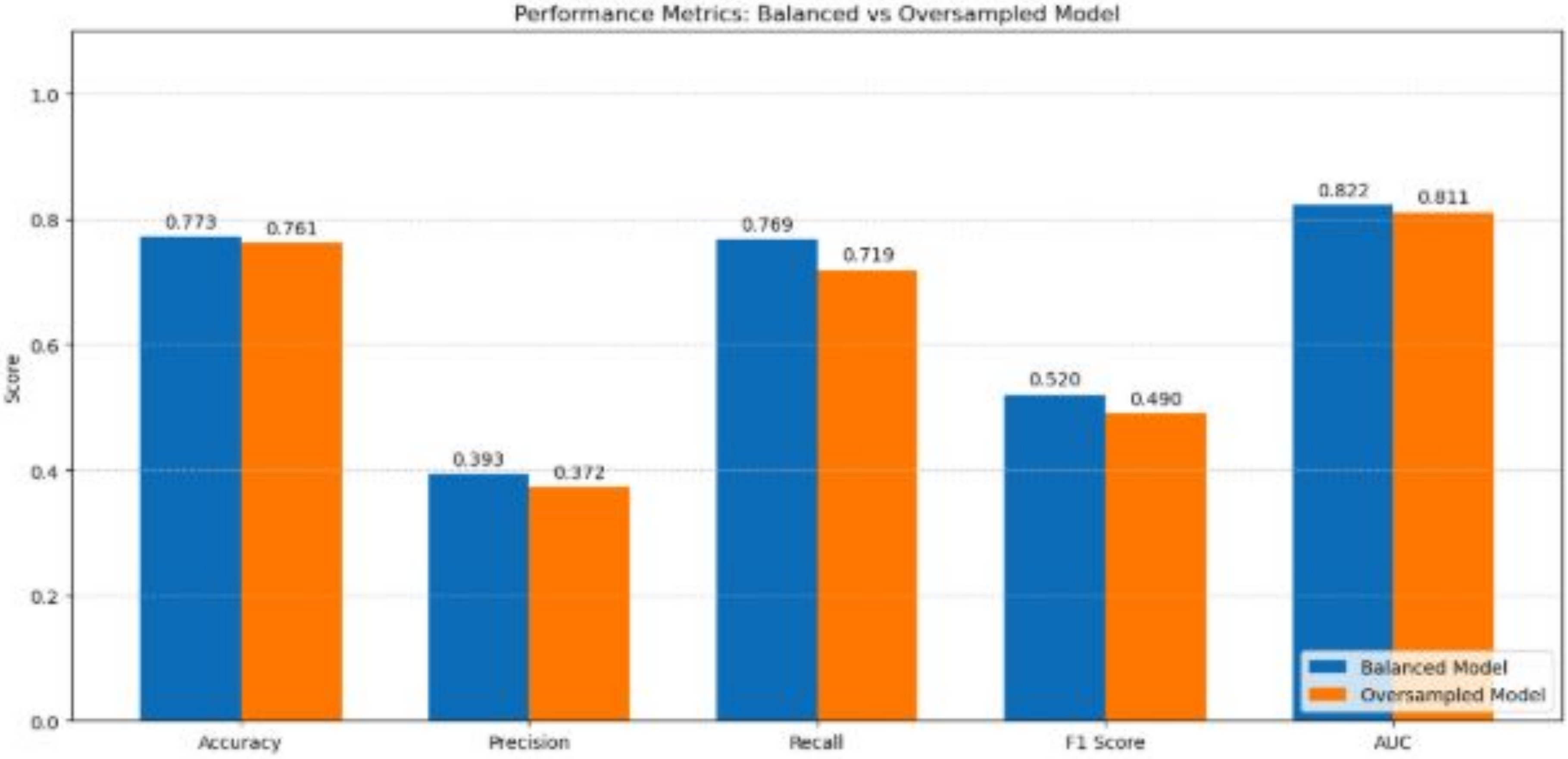
Logistic Regression

- Identified class imbalance in the dataset
- Created models using two techniques: SMOTE and Class weights
- Identified the best one

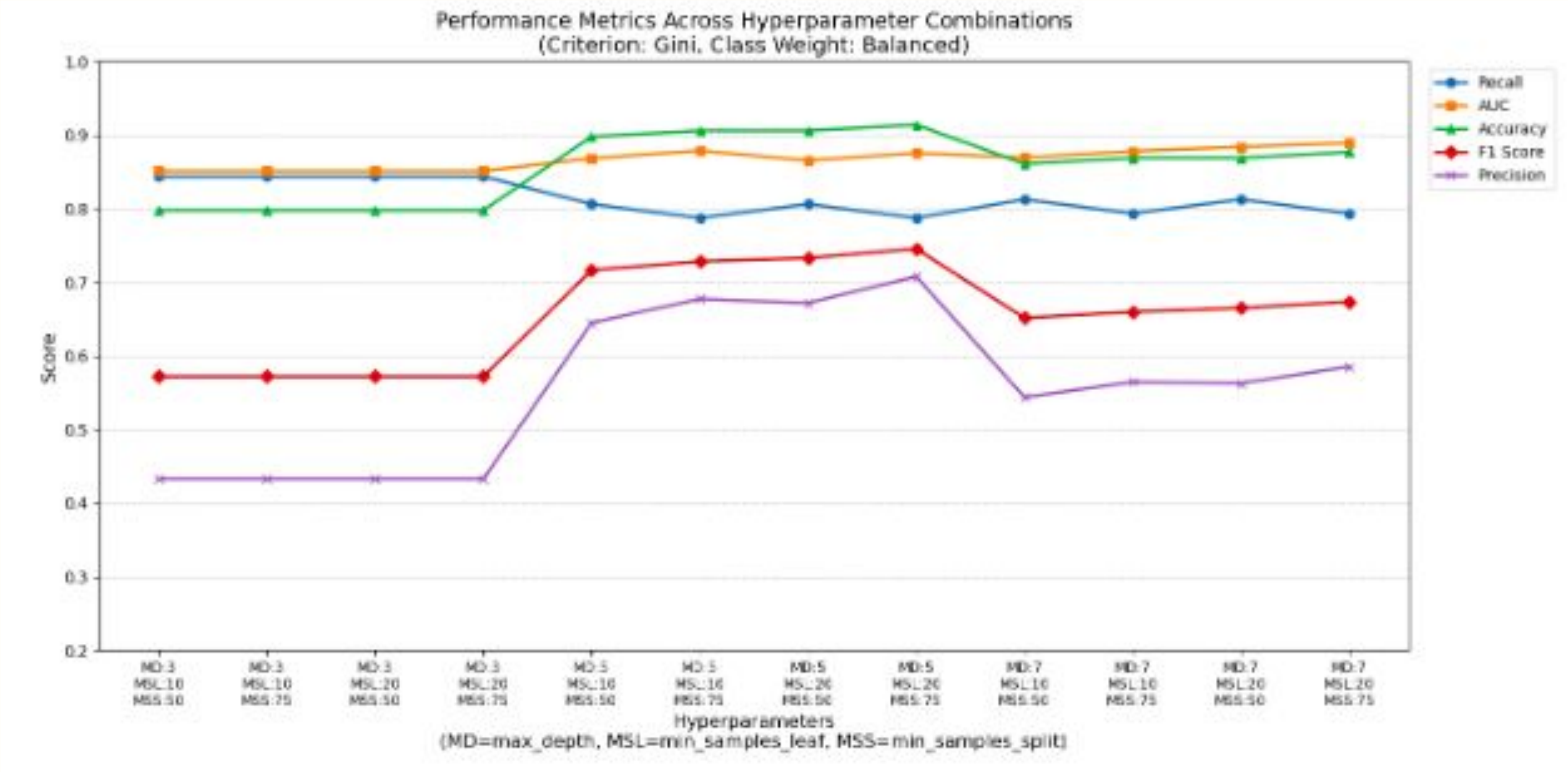
Decision Tree

- Started by creating a vanilla model
- Did hyperparameter tuning and pruning to improve performance
- Identified best hyperparameters and built the best decision tree

Logistic Regression Models

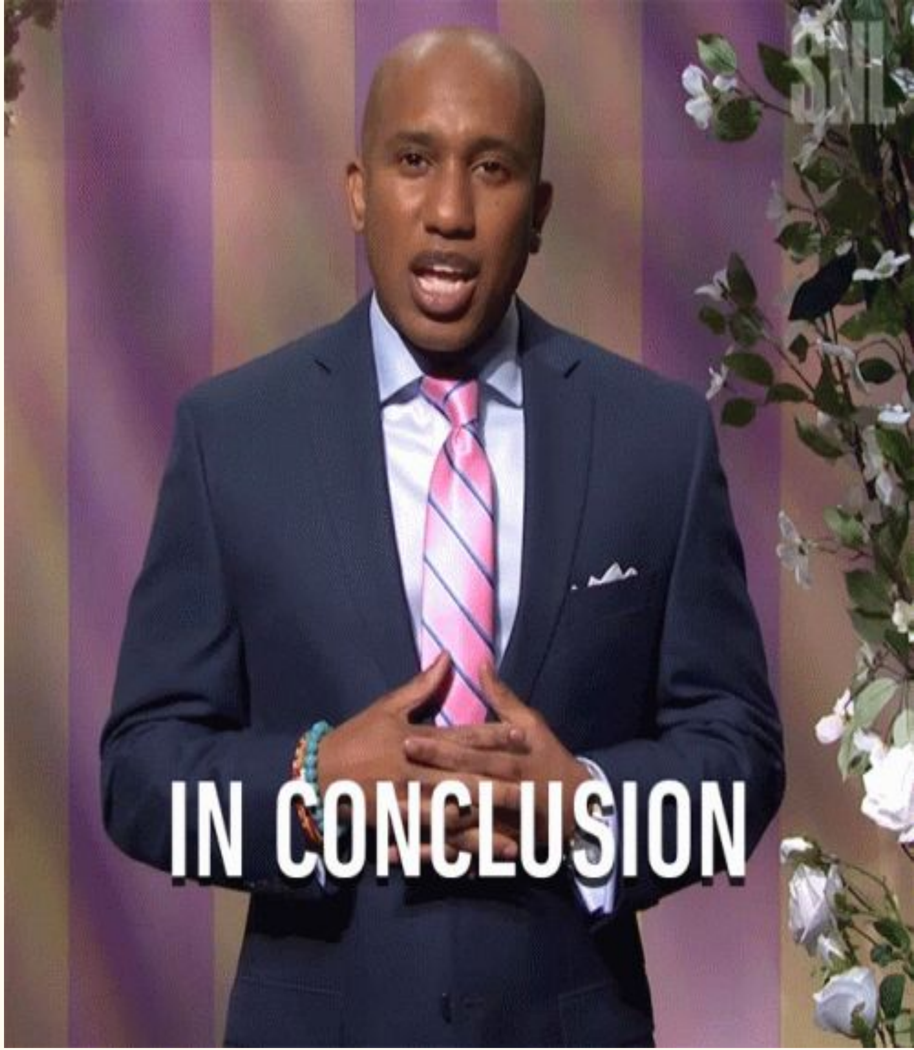


Decision Trees

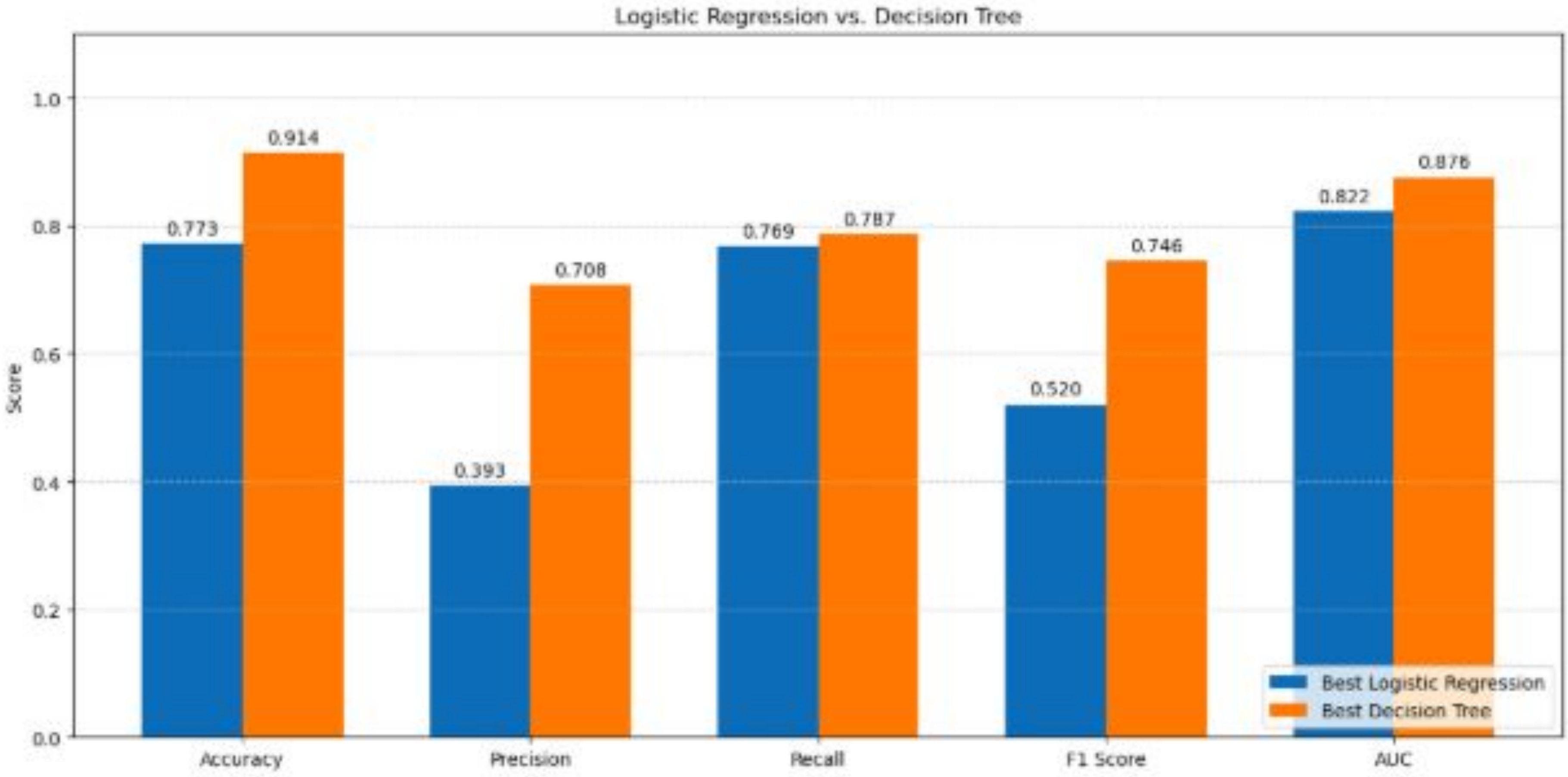


Conclusion

- Voicemail plan, International plan and Number of customer service calls are some of the most significant metrics determining churning.
- Decision tree is the better model over the logistic regression model



Logistic Regression Model vs Decision Tree model



Recommendations

Model Recommendations

- Use the decision tree for **prediction**
- Use the logistic regression model for **inference**

Business Recommendations

- Lower the price of an international plan
- Get more customers to use a voicemail plan
- Improve customer service operations

Next Steps

- Do **cross-validation** to see if performance can be improved further
- Play around with different hyperparameter combinations
- Seek more data on customer churn to create more accurate models

THANK YOU

Author: Ngundo Muithya

Email: ngundolarrymuithya@gmail.com